

Direct quadratic polynomial fitting for fringe peak detection of white light scanning interferograms

Min-Cheol Park

Seung-Woo Kim, MEMBER SPIE

Korea Advanced Institute of Science and Technology

Department of Mechanical Engineering

Science Town, Taejeon, 305-701

Korea

Abstract. We present a new computational method of white light scanning interferometry for 3-D surface mapping. This method accomplishes the task of detecting the true peak of the interference fringe in two steps. The first step is global search locating the envelope peak by fitting sampled intensity data directly to a symmetric quadratic polynomial. The second step is fine-tuning to precisely determine the fringe peak by compensating for the phase shift on reflection using the absolute fringe order identified by the envelope peak obtained in the first step. This two-step method offers an efficient means of computation to provide a good measuring accuracy with high noise immunity owing to its inherent reliance on least squares principles. © 2000 Society of Photo-Optical Instrumentation Engineers. [S0091-3286(00)00204-X]

Subject terms: white light scanning interferometry; surface metrology; phase measuring interferometry; peak detection of interferograms; quadratic polynomial fitting.

Paper 980344 received Sep. 8, 1998; revised manuscript received May 5, 1999, and July 27, 1999; accepted for publication Sep. 10, 1999.

1 Introduction

White light interferometry produces short coherence interferograms whose fringe visibility is narrowly localized in the spatial domain so that the optical path difference between the test and reference beams can be scaled without 2π -ambiguity. During the last two decades since a relevant USA patent was first filed in 1980 by Balasubramanian,¹ much attention has been paid to 3-D surface mapping using white light interferometry. Nowadays quite a few noble techniques are available in the published literature as faithfully reviewed by Creath.² The techniques differ in the intricate way of fringe data processing, but just about all of them rely upon the common tactics of locating the peak of interferograms through intensive computation with aids of fast desktop computers. Kino and Chim^{3,4} first used the Fourier and later Hilbert transforms to extract the visibility envelope function in the frequency domain and then inversely transform the result back in the spatial domain to determine the envelope peak. Caber and Cohen et al.^{5,6} proposed a fast demodulation method isolating the visibility envelope function by signal filtering. Groot and Deck⁷⁻⁹ revisited the Fourier transform as previously but succeeded in locating the fringe peak directly in the frequency domain, easing sampling requirements and computational burden. There were some other types of attempts made to modify the well-established phase measuring algorithms of coherent interferometry to be suited for white light interferometry. For example, Larkin¹⁰ proposed an adaptive non-linear algorithm of five-buckets, and Sandoz and Sandoz et al.^{11,12} suggested a seven-bucket phase measuring algorithm. As a new alternative solution to the fringe peak detection, Sandoz¹³ investigated the possibility of adopting the wavelet transform method.

The previous work may be distinguished into two groups, depending upon which distinctive point they pursue as the peak of interferograms. The first one seeks the summit of the envelope profile of the sampled interference intensity, which is hereafter termed the envelope peak.^{1,3-6} The second one identifies the fringe peak that corresponds to the maximum crest of the interference intensity varying in a sinusoidal pattern.⁷⁻¹³ The fringe peak is more sensitive to the optical path difference than the envelope peak, thus the approaches of the second group have the potential of providing better measurement accuracy. However, locating the fringe peak requires phase measurement often disturbed by the 2π -ambiguity in identifying absolute fringe order. Now in this paper, we propose a new computational method that accomplishes the task of detecting the fringe peak in two steps. The first step performs global search locating the envelope peak by fitting sampled intensity data directly to a symmetric quadratic polynomial. The second step then implements fine adjustment to precisely determine the fringe peak by compensating for the phase shift on reflection using the absolute fringe order identified in the first step. This two-step method offers a significant reduction in the required computation time and also provides a good measuring accuracy with high noise immunity owing to its inherent reliance on least squares principles. Hereafter our peak detection method will be described in detail, followed by a series of computational simulation and experiments to verify its effectiveness of sampling requirements, noise immunity, calculation efficiency and accuracy.

2 Detection of Envelope Peak

White light scanning interferometry for 3-D surface metrology can be performed with a variety of optical setups such

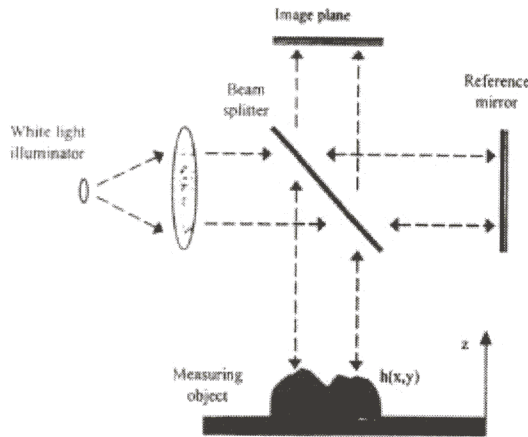


Fig. 1 Basic configuration of white light scanning interferometry.

as Michelson, Mirau, and Linnik interferometers. Here we take a Michelson configuration shown in Figure 1 in its simplest form to explain the working principles of the peak detection algorithm to be proposed. A two-dimensional photodetector array samples scanning interferograms while moving the target surface along the z direction. The height $h(x,y)$ of a point $P(x,y)$ on the surface is measured from the reference point of $z=0$ where scanning starts for intensity data sampling. The corresponding interferogram undergoes the intensity variation of reference¹⁴

$$I(z) = I_B [1 + g(h-z) \cos\{2k_c(h-z) + \alpha\}]. \quad (1)$$

In Eq. (1), the notation (x,y) is omitted for convenience, while $g(h-z)$ is the visibility envelope function; I_B the background intensity; k_c the wavenumber decided by the central wavelength of light source; and α the phase change on reflection. The visibility envelope function $g(h-z)$ is affected by many optical characteristics such as the spectral distribution of the light source, the numerical aperture of the objective, and the phase change on reflection from the surface and the reference¹⁵. If the effect of the numerical aperture is neglected, $g(h-z)$ may be assumed as a monotonously convex function of Gaussian or sinc type for most practically available light sources.¹⁴ In this case, $g(h-z)$ holds symmetry about its global maximum point $z=h$, where the envelope peak occurs. In general, because of the phase change on reflection, the envelope peak does not coincide with the fringe peak that represents the actual peak of the fringe intensity varying in a sinusoidal pattern. As illustrated in Figure 2, the two peaks have a relative offset of

$$\Delta h = h - h^* = -\frac{\alpha}{2k_c}, \quad (2)$$

where h represents the envelope peak and h^* the fringe peak.

The 3-D surface mapping requires identifying the relative position of every single interferogram in the spatial domain. This task may be performed by locating either the

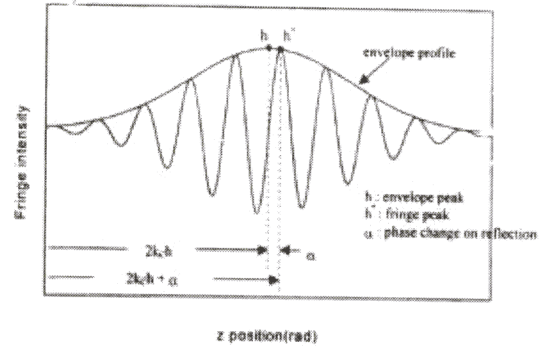


Fig. 2 The envelope peak vs. fringe peak. The abscissa is expressed in terms of radians corresponding to scanning position.

envelope peak h or the fringe peak h^* . For this purpose, the visibility envelope function is approximated by the quadratic polynomial so that Eq. (1) is rewritten as

$$I(z) = I_B + I_B [a_1 + a_2 z + a_3 z^2] \cos(2k_c(h-z) + \alpha). \quad (3)$$

In Eq. (3), a_1 , a_2 , and a_3 are the coefficients that determine the envelope peak as $h = -a_2/2a_3$. Now, let $I(z_i)$ be the intensity value discretely sampled during scanning at $z = z_i$ for $i = 1, 2, \dots, N$, then Eq. (3) is rearranged as

$$I(z_i) = I_B + I_B (a_1 + a_2 \delta_i + a_3 \delta_i^2) (\cos \delta_i \cos \theta + \sin \delta_i \sin \theta). \quad (4)$$

Two new phase variables are defined in Eq. (4) as $\delta_i = 2k_c z_i$ and $\theta = 2k_c h + \alpha$. The former is related to the scanning position, while the latter represents the phase of the interferometric intensity at the fringe peak, i.e.

$$h^* = \frac{\theta}{2k_c}. \quad (5)$$

Eq. (4) indicates that besides the coefficients a_1 , a_2 , and a_3 , there are two other unknowns of I_B and θ to be identified. Least squares approach is adopted to fit the five unknowns simultaneously using sampled intensity data. For this, the error function to be minimized is formulated as

$$\begin{aligned} E &= \sum_{i=1}^N [I_i^* - I(z_i)]^2 \\ &= \sum_{i=1}^N [I_i^* - I_B - I_B (a_1 + a_2 \delta_i + a_3 \delta_i^2) \\ &\quad \times (\cos \delta_i \cos \theta + \sin \delta_i \sin \theta)]^2, \end{aligned} \quad (6)$$

where I_i^* denotes the actually measured value of $I(z_i)$. Necessary conditions for this least squares fitting are given as

$$\frac{\partial E}{\partial I_B} = \frac{\partial E}{\partial a_1} = \frac{\partial E}{\partial a_2} = \frac{\partial E}{\partial a_3} = \frac{\partial E}{\partial \theta} = 0. \quad (7)$$

The above five simultaneous equations are highly nonlinear in terms of the independent unknowns, thus their solutions are not obtained in a straightforward manner. One way out is to resort to numerical techniques, but this is not desirable because of heavy computational burden. An alternative way to be adopted hereafter is introducing a new set of unknowns by combining the original ones so that their corresponding necessary conditions become linear. One such substitutive set of new unknowns found for the purpose is given in a column vector of

$$C = [c_1 \ c_2 \ c_3 \ c_4 \ c_5 \ c_6 \ c_7]^T$$

$$= [I_B \ a_1 I_B \cos \theta \ a_2 I_B \cos \theta \ a_3 I_B \cos \theta$$

$$\times a_1 I_B \sin \theta \ a_2 I_B \sin \theta \ a_3 I_B \sin \theta]^T. \quad (8)$$

Then necessary conditions are reconfigured as

$$\frac{\partial E}{\partial c_1} = \frac{\partial E}{\partial c_2} = \frac{\partial E}{\partial c_3} = \frac{\partial E}{\partial c_4} = \frac{\partial E}{\partial c_5} = \frac{\partial E}{\partial c_6} = \frac{\partial E}{\partial c_7} = 0. \quad (9)$$

Assuming that the total number of sampled intensity data is larger than 7, i.e., $N > 7$, Eq. (9) can rigorously be worked out in the form of linear matrix equation of

$$A(\delta_i) \cdot C = B(I_i) \quad \text{or} \quad C = A^{-1}(\delta_i) \cdot B(I_i). \quad (10)$$

The matrices A and B in the above equation are given in terms of the sampled data of δ_i and I_B as

$$A(\delta_i) = \begin{bmatrix} N & \sum \cos \delta_i & \sum \delta_i \cos \delta_i & \sum \delta_i^2 \cos \delta_i & \sum \sin \delta_i & \sum \delta_i \sin \delta_i & \sum \delta_i^2 \sin \delta_i \\ \sum \cos \delta_i & \sum \cos^2 \delta_i & \sum \delta_i \cos^2 \delta_i & \sum \delta_i^2 \cos^2 \delta_i & \sum \sin \delta_i \cos \delta_i & \sum \delta_i \sin \delta_i \cos \delta_i & \sum \delta_i^2 \sin \delta_i \cos \delta_i \\ \sum \delta_i \cos \delta_i & \sum \delta_i \cos^2 \delta_i & \sum \delta_i^2 \cos^2 \delta_i & \sum \delta_i^3 \cos^2 \delta_i & \sum \delta_i \sin \delta_i \cos \delta_i & \sum \delta_i^2 \sin \delta_i \cos \delta_i & \sum \delta_i^3 \sin \delta_i \cos \delta_i \\ \sum \delta_i^2 \cos \delta_i & \sum \delta_i^2 \cos^2 \delta_i & \sum \delta_i^3 \cos^2 \delta_i & \sum \delta_i^4 \cos^2 \delta_i & \sum \delta_i^2 \sin \delta_i \cos \delta_i & \sum \delta_i^3 \sin \delta_i \cos \delta_i & \sum \delta_i^4 \sin \delta_i \cos \delta_i \\ \sum \sin \delta_i & \sum \sin \delta_i \cos \delta_i & \sum \delta_i \sin \delta_i \cos \delta_i & \sum \delta_i^2 \sin \delta_i \cos \delta_i & \sum \sin^2 \delta_i & \sum \delta_i \sin^2 \delta_i & \sum \delta_i^2 \sin^2 \delta_i \\ \sum \delta_i \sin \delta_i & \sum \delta_i \sin \delta_i \cos \delta_i & \sum \delta_i^2 \sin \delta_i \cos \delta_i & \sum \delta_i^3 \sin \delta_i \cos \delta_i & \sum \delta_i \sin^2 \delta_i & \sum \delta_i^2 \sin^2 \delta_i & \sum \delta_i^3 \sin^2 \delta_i \\ \sum \delta_i^2 \sin \delta_i & \sum \delta_i^2 \sin \delta_i \cos \delta_i & \sum \delta_i^3 \sin \delta_i \cos \delta_i & \sum \delta_i^4 \sin \delta_i \cos \delta_i & \sum \delta_i^2 \sin^2 \delta_i & \sum \delta_i^3 \sin^2 \delta_i & \sum \delta_i^4 \sin^2 \delta_i \end{bmatrix} \quad (11)$$

and

$$B(I_i) = \left[\sum I_i \ \sum I_i \cos \delta_i \ \sum I_i \delta_i \cos \delta_i \ \sum I_i \delta_i^2 \cos \delta_i \ \sum I_i \sin \delta_i \ \sum I_i \delta_i \sin \delta_i \ \sum I_i \delta_i^2 \sin \delta_i \right]^T. \quad (12)$$

where the notation Σ means $\sum_{i=1}^N$. Once all the elements of A and B are prepared with sampled data, the unknown matrix C are readily solved from Eq. (10). Then the global maximum of the quadratic polynomial corresponding to the envelope peak is finally obtained as

$$h = \frac{1}{2k_c} \left(-\frac{a_2}{2a_3} \right) = \frac{1}{2k_c} \left(-\frac{c_3}{2c_4} \right) \quad \text{or} \quad h = \frac{1}{2k_c} \left(-\frac{c_6}{2c_7} \right). \quad (13)$$

One important parameter that should be carefully decided in implementing the above algorithm is the sampling interval, which is defined as $\Delta \delta_i = 2k_c \Delta z_i$ where $\Delta z_i = z_i - z_{i-1}$. The aforementioned least square scheme does not

require that $\Delta \delta_i$ be a specific number, as it works with any sized sampling intervals. Nevertheless, from the viewpoint of computational efficiency, it is advantageous to satisfy the equal spacing condition of

$$\Delta \delta_i = \delta_i - \delta_{i-1} = \frac{2\pi}{M} \quad i = 1, \dots, N, \quad (14)$$

where the divider constant M is an integer of either 3 or 4. The use of equally spaced sampling interval removes the need of computing time-consuming cosine and sine functions appearing in the matrices A and B . For example, if M is 4 (i.e., $\Delta \delta_i$ equals $\pi/2$), then

$$A = \begin{bmatrix} N & 0 & \sum (-1)^{j-1} \delta_{2j-1} & \sum (-1)^{j-1} \delta_{2j-1}^2 & 1 & \sum (-1)^{j-1} \delta_{2j} & \sum (-1)^{j-1} \delta_{2j}^2 \\ 0 & N/2 & \sum \delta_{2j-1} & \sum \delta_{2j-1}^2 & 0 & 0 & 0 \\ \sum (-1)^{j-1} \delta_{2j-1} & \sum \delta_{2j-1} & \sum \delta_{2j-1}^2 & \sum \delta_{2j-1}^3 & 0 & 0 & 0 \\ \sum (-1)^{j-1} \delta_{2j-1}^2 & \sum \delta_{2j-1}^2 & \sum \delta_{2j-1}^3 & \sum \delta_{2j-1}^4 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & N/2 & \sum \delta_{2j} & \sum \delta_{2j}^2 \\ \sum (-1)^{j-1} \delta_{2j} & 0 & 0 & 0 & \sum \delta_{2j} & \sum \delta_{2j}^2 & \sum \delta_{2j}^3 \\ \sum (-1)^{j-1} \delta_{2j}^2 & 0 & 0 & 0 & \sum \delta_{2j}^2 & \sum \delta_{2j}^3 & \sum \delta_{2j}^4 \end{bmatrix} \quad (15)$$

and

$$B = \left[\sum I_i \sum (-1)^{j-1} I_{2j-1} \sum (-1)^{j-1} I_{2j-1} z_{2j-1} \sum (-1)^{j-1} I_{2j-1} z_{2j-1}^2 \sum (-1)^{j-1} I_{2j} \sum (-1)^{j-1} I_{2j} z_{2j} \sum (-1)^{j-1} I_{2j} z_{2j}^2 \right]^T, \quad (16)$$

where the notation Σ means $\sum_{i=1}^{N/2}$ this time. The matrix inversion for A^{-1} in Eq. (10) needs to be done only once and the result is repeatedly applied to all the surface points. In doing that, the total number of intensity data N needs to be selected large enough for the data to be well distributed over the limited coherence length of interferograms. Finally, it is worthwhile to note that the results of Eqs. (15) and (16) may be regarded as an extended version of the least squares algorithm of the phase shifting technique proposed by Greivenkamp.¹⁶ The previous one aims at the phase measurement of coherent interferometry, while the new one presented in this paper is intended especially for fitting short coherent white light interferograms.

3 Detection of Fringe Peak

Now we begin with the second step of our algorithm. As explained in the introductory section of this paper, the fringe peak is more desirable to trace than the envelope peak. The main reason is that the fringe intensity itself yields relatively higher sensitivity to the path difference than the envelope profile. Another reason is that the envelope peak located in the first step by fitting the quadratic polynomial has some errors because of the asymmetry actually observed in the sampled fringe intensity pattern. Thus, this second step devotes itself to locating the fringe peak. Recall that in Eq. (4) the phase θ was defined to be $\theta = 2k_c h + \alpha$, which in fact denotes the total phase accumulated from the reference point of $z=0$ to the fringe peak, i.e., $h^* = \theta/2k_c$. The phase θ can explicitly be described using the solution matrix C of Eq. (10) determined in the first step such that

$$\theta = 2\pi n + \theta_m, \text{ where } \theta_m = \arctan\left(\frac{\sin \theta}{\cos \theta}\right) = \arctan\left(\frac{c_3}{c_2}\right). \quad (17)$$

Note that θ_m is the principal value limited in the range of $-\pi$ to π as the output of the arctangent operation. Therefore, the absolute fringe order n should be identified to determine the phase θ without 2π -ambiguity. The explicit expression of the absolute fringe order is derived by substituting the definition of $\theta = 2k_c h + \alpha$ into Eq. (17) as

$$n = \frac{1}{2\pi} [2k_c h + \alpha - \theta_m]. \quad (18)$$

The values of h and θ_m of the above equation are known at this stage of computation. However, this is not the case for the phase change α , which is affected in a complex way by the kind of target materials and the numerical aperture of the microscope objective.¹⁵ Nevertheless, the precise value of α for a given target material may be obtained through carefully prepared calibration,¹⁷ so the absolute fringe order can be determined using Eq. (18). If the phase change α is simply assumed constant, then the fringe peak may be expressed as

$$h^* = \frac{1}{2k_c} \left(\theta_m + 2\pi \cdot \text{Int} \left[\frac{2k_c h - \theta_m}{2\pi} \right] \right), \quad (19)$$

where the function $\text{Int}[\cdot]$ outputs the closest integer value to its argument as the absolute order n . In this case, as illustrated in Figure 3, the absolute fringe order of the fringe peak is assumed the same as that of the envelope

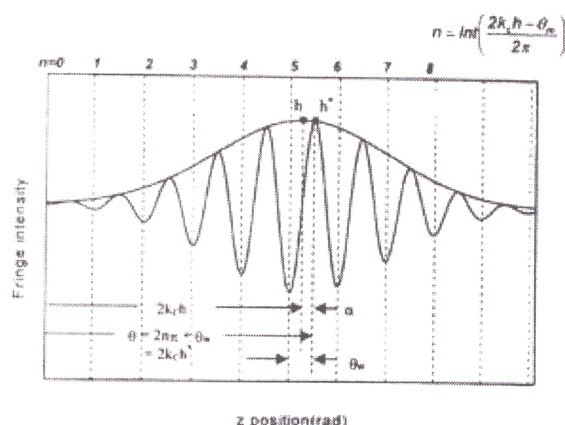


Fig. 3 The absolute order of the fringe peak.

peak. This assumption may cause measurement errors when the target surface is composed of multiple materials with different values of α . On the other hand, for the target of a single material, no measurement errors are induced, since the assumption of constant α simply results in a uniform offset in the measured profile. It should also be noted that the $\text{Int}()$ function in Eq. (19) may be susceptible to noise if the argument is close to the middle of two consecutive integers. In this case, the true value of the fringe order may be readily corrected by observing the difference $h-h^*$.

In completing the description of the proposed algorithm, the computational efficiency is discussed by counting the required total number of multiplications. Our two-step algorithm demands $3N$ multiplications for forming the matrix B of Eq. (16) and additionally 32 multiplications and one arctangent operation for computation of $A^{-1}B$, h , and θ . Consequently, the total number of multiplications makes $3N+33$. This figure is compared with those of existing techniques as listed in Table 1. The contents are quoted from the comparison analysis rigorously performed by Larkin¹⁰ as for the Hilbert transform by Chim and Kino,⁴ the frequency domain analysis by Deck and de Groot,⁸ and the adaptive nonlinear algorithm by Larkin himself.¹⁰ Included also in the table are the seven-bucket phase measuring and wavelet transform algorithms proposed by Sandoz and Sandoz et al.¹¹⁻¹³

Table 1 Comparison of computational burdens of existing algorithms.

	Number of multiplications
Adaptive nonlinear algorithm	$2N+29$
Frequency domain processing	$N(\log_2 N+3)$
Hilbert transform	$6N$
Quadratic polynomial fitting	$3N+33$
Seven-bucket phase measuring algorithm	$20N(N=3)$
Wavelet transform	$25N$

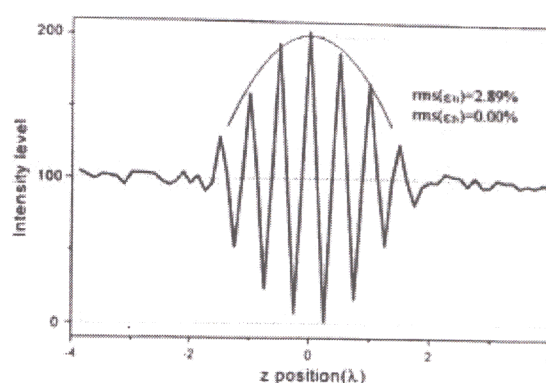


Fig. 4 An exemplary intensity profile generated for computer simulation.

4 Algorithm Evaluation

A series of computer simulations were performed to test the proposed two-step algorithm with regard to measurement accuracy and noise immunity. For quantitative evaluation, a set of discrete intensity data was artificially prepared using a representative expression of the white light interferogram of

$$I_i = \text{Int} \left\{ 100 + 100 \frac{\sin(2k(h-z_i))}{(2k(h-z_i))} \times \cos(2k(h-z_i) + \alpha + \epsilon_{2i}) + \epsilon_{1i} \right\}. \quad (20)$$

The above formula includes the random variables ϵ_{1i} and ϵ_{2i} whose root-mean-square values are adjusted to simulate the degree of external disturbances. The latter variable represents the scanning error of the piezoelectric actuator, while the former the vibration and electrical noise encountered in measurement. The function $\text{Int}(x)$ models the analog-to-digital quantizing process of intensity sampling. Assuming 8-bit quantizing, the sampled intensity values were scaled to lie in the range of 0 to 255, while their mean and visibility amplitude are fixed at 100 and 100, respectively. The total number of sampled data N was set to be 21 with the sampling interval of $\Delta z_i = \pi/2$. Figure 4 shows a typical profile of intensity data generated for computer simulation.

First, the accuracy of the proposed algorithm was evaluated against a test surface of Figure 5(a), which is ideally flat with a tilt of $\lambda/2$ with the optical axis. A total of 40 representative points were selected along the test surface and their heights were computed using the intensity data generated by Eq. (20). Effects of ϵ_{1i} and ϵ_{2i} were not considered in this particular simulation. Figure 5(b) shows test results of measurement errors. Curve A presents the offset between the envelope peak h and the true surface height h_{true} , which exhibits a periodic pattern of $\lambda/8$ pitch coinciding with the sampling interval $\Delta z_i = \pi/2$. This offset arises mainly from the approximation of the visibility function by the quadratic polynomial in the proposed algorithm, but its magnitude is not significant being less than a maxi-

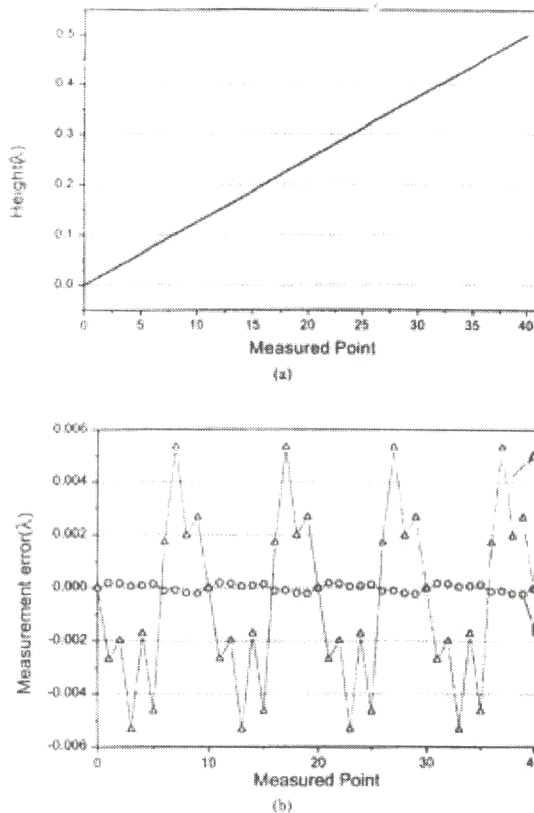


Fig. 5 Simulation results of measurement accuracy: (a) an ideal surface with $\lambda/2$ height tilt; (b) measurement errors (curve A: $h - h_{\text{true}}$, curve B: $h^* - h_{\text{true}}$).

mum of $\pm 0.0053\lambda$. Curve B in the same figure indicates the offset between the fringe peak h^* and the true height h_{true} , which lies within the range of $\pm 0.0002\lambda$. This simulation confirms that the proposed algorithm is reasonably reliable and that detecting the fringe peak provides better accuracy in comparison with that of the envelope peak.

Figure 6 shows another simulation result regarding the robustness of the proposed algorithm against random external disturbances. The upper figure reveals how the measurement accuracy is affected as the root-mean-squared level of the variable ϵ_{1i} increases from 0 to 14.2% of the visibility amplitude. It is seen that measurement errors remain less than 0.01λ in the disturbance range of 0 to 1.5% and gradually increase to 0.05λ when the level of ϵ_{1i} reaches 8%. This result is found quite satisfactory since the level of ϵ_{1i} encountered in actual measurement is usually about 2%, at worst 5%. Now turning to PZT stepping errors, Figure 6(b) shows influences of the disturbance ϵ_{2i} . The result indicates that the induced error is negligible in the range of 0~4.0%, and we conclude that the proposed algorithm has a good immunity to random external disturbances owing to the averaging process of the least squares technique adopted.

Now, the validity of approximating the visibility enve-

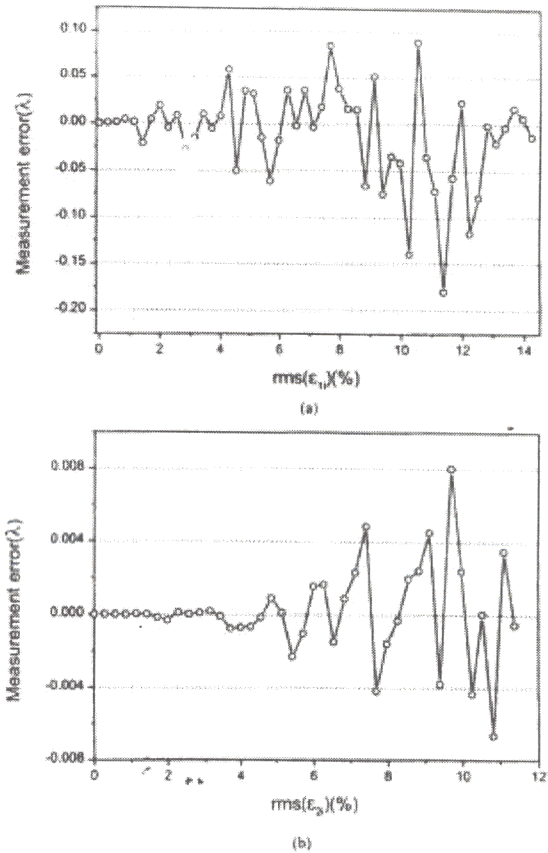


Fig. 6 Simulation results of measurement errors against the external disturbances ϵ_{1i} and ϵ_{2i} : (a) measurement errors versus ϵ_{1i} ; (b) measurement errors versus ϵ_{2i} .

lope function as the quadratic polynomial was verified by using an actual interferogram discretely sampled in Figure 7(a), which shows a certain degree of asymmetry. The true values of the envelope peak and fringe peak were found accurately by fitting the 6th-order polynomial and compared with the results of the quadratic polynomial in our algorithm. The comparison results are shown in Figure 7(b), which were made over the tilted test surface of Figure 4(a). In the case of the envelope peak, the difference has a variation of 26.09 ± 29.22 nm over the entire surface, while the phase peak has a relatively very small variation of -0.35 ± 0.62 nm. This concludes that the quadratic polynomial provides a good approximation even for real intensity profiles and, more importantly, measurement errors can safely be neglected in detecting the fringe peak.

Now an experimental result is shown in Figure 8, which was performed against an optical flat with a $20\times$ Mirau objective of 0.4 NA value. The number of sampled data N was 21 with a scanning interval of $\lambda/8$. Its mean roughness is found to be $R_q = 0.4506$ nm, which demonstrates the proposed algorithm provides good measuring capability. Another experimental result is shown as Figure 9, which measures the height profile of a step height made of dissimilar

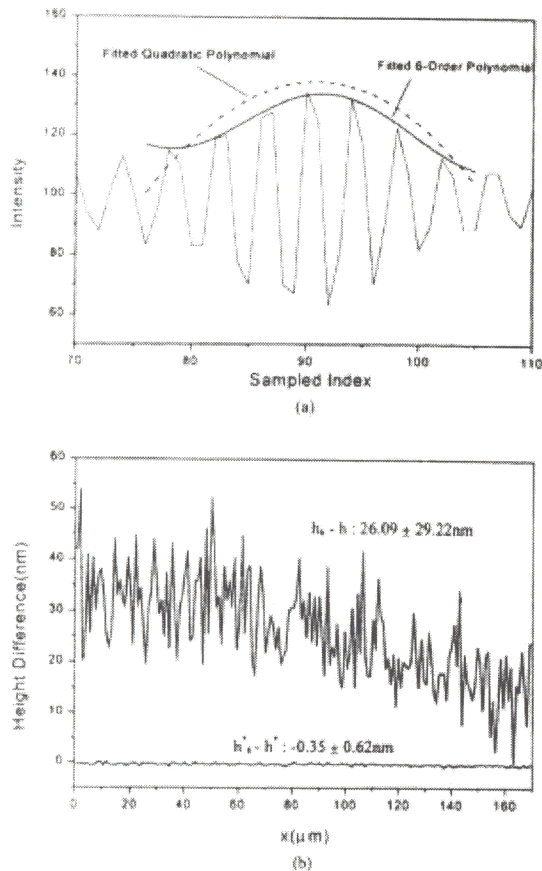


Fig. 7 Validity tests of assuming the visibility function as a quadratic polynomial: (a) intensity profile of actual interferogram; (b) comparison of detection errors.

metallic materials. The measured profiles of h and h^* have a significant offset of 90 nm. The two-peak points h and h^* are differently shifted from its origin depending on the refractive indices, the source wavelength, and the NA value

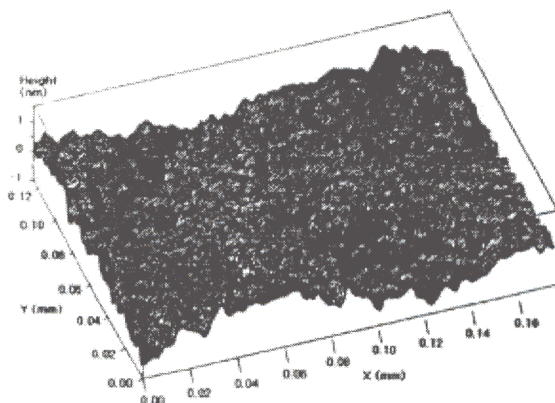


Fig. 8 Reconstructed surface profile of an optical flat ($R_q = 0.4457$ nm).

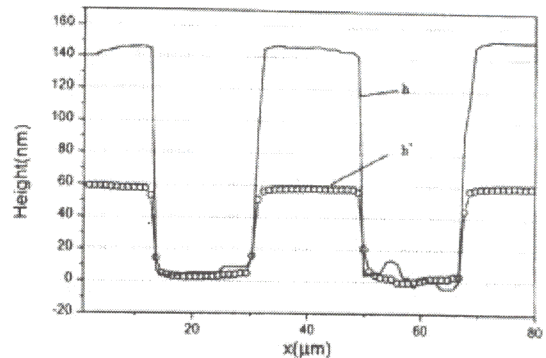


Fig. 9 Cross sectional profile of a step height with two different metals.

of the objective. Thus, at the current stage of our research, it is not possible to determine the true heights of dissimilar metallic materials. Further investigation is underway to compensate for different values of phase changes on measurement accuracy.

5 Conclusions

A new peak detection algorithm has been proposed and tested for automated 3-D surface mapping using white light scanning interferometry. By assuming the visibility function of white light interferograms as a quadratic polynomial, the envelope peak is searched so as to minimize the error sum between the measured intensity data and the analytical intensity. Then the fringe peak is located through a fine-tuning process for better measurement accuracy. This new algorithm provides an efficient means of computation to determine the envelope peak simply with a single-step matrix multiplication. In addition, good robustness is obtained against external random disturbances on measured intensities since the algorithm is based upon least squares principles. Computer simulation reveals that the maximum measurement error is less than $\pm 0.05\lambda$ in the presence of external disturbances.

References

1. N. Balsubramanian, "Optical system for surface topography measurement," U.S. Patent No. 4,340,306 (1980).
2. K. Creath, "Sampling requirements for white light interferometry," *Proc. Fringe '97*, Academic Verlag, 52-59 (1997).
3. G. Kino and S. Chim, "The Mirau correlation microscope," *Appl. Opt.* 29(26), 3775-3783 (1990).
4. S. Chim and G. Kino, "Three-dimensional image realization in interference microscopy," *Appl. Opt.* 31(14), 2550-2553 (1992).
5. P. Caber, "Interferometric profiler for rough surfaces," *Appl. Opt.* 32(19), 3438-3441 (1993).
6. D. Cohen, P. Caber, and C. Brophy, "Rough surface profiler and method," U.S. Patent No. 5,133,601 (1992).
7. P. de Groot and L. Deck, "Three-dimensional imaging by sub-Nyquist sampling of white-light interferograms," *Opt. Lett.* 18(17), 1462-1464 (1993).
8. L. Deck and P. de Groot, "High-speed noncontact profiler based on scanning white-light interferometry," *Appl. Opt.* 33(31), 7334-7338 (1994).
9. P. Groot and L. Deck, "Surface profiling by analysis of white-light interferograms in the spatial frequency," *J. Mod. Opt.* 42(2), 389-401 (1995).
10. K. Larkin, "Efficient nonlinear algorithm for envelope detection in white light interferometry," *J. Opt. Soc. Am. A* 13(4), 832-843 (1996).

11. P. Sandoz, "An algorithm for profilometry by white-light phase shifting interferometry," *J. Mod. Opt.* 43(8), 1545-1554 (1996).
12. P. Sandoz, R. Devillers, and A. Plata, "Unambiguous profilometry by fringe-order identification in white-light phase-shifting interferometry," *J. Mod. Opt.* 44(3), 519-534 (1997).
13. P. Sandoz, "Wavelet transform as a processing tool in white-light interferometry," *Opt. Lett.* 22(14), 1065-1067 (1997).
14. M. Born and E. Wolf, *Principles of Optics*, Pergamon Press, Sixth ed., 316-323.
15. J. F. Biegen, "Determination of the phase change on reflection from two-beam interference," *Opt. Lett.* 19(21), 1690-1692 (1994).
16. J. Greivenkamp, "Generalized data reduction for heterodyne interferometry," *Opt. Eng.* 23(4), 350-352 (1984).
17. T. Doi, K. Toyoda, and Y. Tanimura, "Effects of phase changes on reflection and their wavelength dependence in optical profilometry," *Appl. Opt.* 36(28), 7157-7161 (1997).



Min-Cheol Park received his MSc degree in mechanical engineering from Korea Advanced Institute of Science and Technology (KAIST) in 1996 and is currently pursuing his PhD degree in mechanical engineering. His research interests include 3D-profile measurement with WSI and 2D-feature measurement with optical inspection.



Seung-Woo Kim received his MS in mechanical engineering from Korea Advanced Institute of Science and Technology (KAIST) and his PhD in precision machine system design from Cranfield University, United Kingdom. He is currently working as a professor in the Department of Mechanical Engineering at KAIST and is in charge of the Laboratory of Precision Engineering and Metrology. His research interest includes precision machine systems design, mechatronics, and optical metrology. He is a member of SPIE, OSA, ASME, and SME.